Larger initial overlaps for the log-balanced study

same Hamiltonians, different initial states

In short

1. **The small overlaps come from how the initial states are drawn, not from the Hamiltonians.** The study draws them uniformly from every basis state with $p > 10^{-12}$. Most of those states lie in the exponentially small tail of the ground state, so the median overlap is $10^{-6.6}$ and 55% of the runs start below 10^{-6}.
2. **The current overlap is tangled with the density** (correlation 0.59 between $\log_{10}$ overlap and $\log_{10}$ density). A low-density ground state has a few heavy states and a huge light tail, and the uniform draw lands in the tail. The "by overlap" plots are therefore partly "by density" plots.
3. **Recommendation:** draw the initial states uniformly from the states the density counts, $p > 0.1/2^N$. This is a one-line change in `run_cell`. The median overlap rises to $10^{-2.7}$, no run starts below $0.1/2^N$, and the overlap becomes independent of the density (correlation -0.07). The Hamiltonians are unchanged.
4. **The price:** a ground state with fewer than six such states gets fewer initial states. This affects 715 of the 2880 Hamiltonians, all with density below $10^{-1.1}$, and gives 14 491 instead of 17 280 runs. The dropped runs are exactly the ones that started around 10^{-9}.

1 Where the overlaps come from

I used the 2880 Hamiltonians the production run selected (`balanced_hamiltonians_log.csv` in the shard directory): 20 per $(N, \text{dimension}, \Delta, \text{max distance})$. I solved every ground state once and drew six initial states per Hamiltonian under several schemes (`initial_states.py`). The scheme *current* uses the same generator and the same call as `run_cell`. It reproduces 17 274 of the 17 280 production runs state by state, so the numbers below describe the data in the shards.

The current draw spreads the overlap almost evenly in $\log_{10}$ from 10^{-12} up to about 10^{-3} (Fig. 1, top row). The ground state has some weight on nearly every basis state of its symmetry sector, but almost all of that weight sits on a few of them. The remaining states fill the decades below. A uniform pick from the support is therefore almost always a tail state.

Fig. 2 (top row) shows the consequence. The lower the density, the more tail there is to draw from. At $\log_{10}$ density < -2, 86% of the initial states start below 10^{-6}, although the same Hamiltonians offer states with $p \approx 1$ (grey).

2 Alternatives

All schemes keep the Hamiltonians and change only which six basis states are started from.

scheme	runs	q_5	median	q_{95}	$> 10^{-2}$	$> 10^{-3}$	$< 10^{-6}$	corr.
current: uniform, $p > 10^{-12}$	17 280	−11.4	−6.6	−1.8	6%	15%	55%	0.59
dense: uniform, $p > 0.1/2^N$	**14 491**	**−4.4**	**−2.7**	**−0.0**	**29%**	**61%**	**0%**	**−0.07**
dense, topped up to six	17 280	−9.9	−2.9	−0.1	25%	51%	13%	0.44
stratified in $\log_{10} p$	16 019	−11.0	−4.8	−0.0	17%	28%	39%	0.15
stratified in $\log_{10} p$, $p > 0.1/2^N$	10 690	−4.0	−2.2	−0.0	45%	75%	0%	−0.28
with probability p (Born)	17 280	−4.7	−2.1	−0.0	47%	73%	1%	0.38
the six largest p (reference)	17 280	−4.6	−1.7	−0.0	61%	79%	1%	0.51

Table 1: Initial overlaps of the 2880 selected Hamiltonians, six initial states each. q_5, median and q_{95} are quantiles of $\log_{10}$ overlap. The last column is the correlation of $\log_{10}$ overlap with $\log_{10}$ density over all runs.

Dense (recommended). It removes the tail and nothing else. The overlap then lies between $0.1/2^N$ and the largest p of the Hamiltonian, and it no longer depends on the density. It uses the same threshold as the density, so "the ground state" means the same set of states in both quantities.

Dense, topped up. It keeps six runs per Hamiltonian, but the top-up states of the low-density Hamiltonians come from the tail again. That brings back 13% of runs below 10^{-6} and the correlation with the density (0.44).

Stratified. It covers the whole range, 10^{-11} to 1, evenly per Hamiltonian, but still puts 39% of the runs below 10^{-6}. Use it only if the tiny overlaps are wanted for their own sake. Restricted to $p > 0.1/2^N$, it leaves too few runs, since strata stay empty.

Born. Drawing with probability p is what measuring the ground state would return, and it gives the largest typical overlaps. It is still correlated with the density (0.38): after the heavy states of a low-density Hamiltonian are taken, the draw without replacement falls into the tail.

3 What sets the range

With the recommended draw, the overlap of a run lies between $0.1/2^N$ and the largest p of its Hamiltonian (Fig. 2, bottom row). Both bounds are physical, so no draw can move them.

- **The lower bound falls with** N, from $10^{-2.8}$ at $N = 6$ to $10^{-4.6}$ at $N = 12$, and the median follows it: $10^{-1.7}$, $10^{-2.4}$, $10^{-3.1}$ and $10^{-3.6}$ at $N = 6, 8, 10, 12$. A dense ground state at $N = 12$ spreads its weight over thousands of states, so each of them carries about $1/2^N$. If overlaps should be compared across N, the relative overlap $p \cdot 2^N$ is the natural axis.
- **The upper bound falls with the density**, because a dense ground state has no heavy state. Three quarters of the overlaps above 10^{-1} come from Hamiltonians with density below 10^{-1}.

4 How to change it

In `EqualNumberOfBitstrings.run_cell`, draw from the states the density counts:

```
# before
non_zero_indices = np.flatnonzero(probabilities > OVERLAP_FLOOR)
# after: the states the ground-state density counts, p > 0.1 / 2^N
non_zero_indices = np.flatnonzero(probabilities > 0.1 / probabilities.size)
```

Everything else can stay.

- The Hamiltonians, the per-cell seeds and with them the SKQD shots are unchanged.
- `load_partial` and `merge` already accept cells with fewer than `num_initial_states` initial states. A cell is checked in blocks of one initial state, so these cells count as finished rather than being redone.
- The run needs a new `SHARD_DIR`. Copy `balanced_hamiltonians_log.csv` into it first: then plan reads the Hamiltonians instead of selecting them again, and they are guaranteed to be the same ones.
- There are 16% fewer runs, so the array finishes sooner.

Reproduce in `StudyOfHamiltonians/`: `python initial_states.py`, then `python overlap_figures.py`, then `pdflatex overlap_report.tex`.

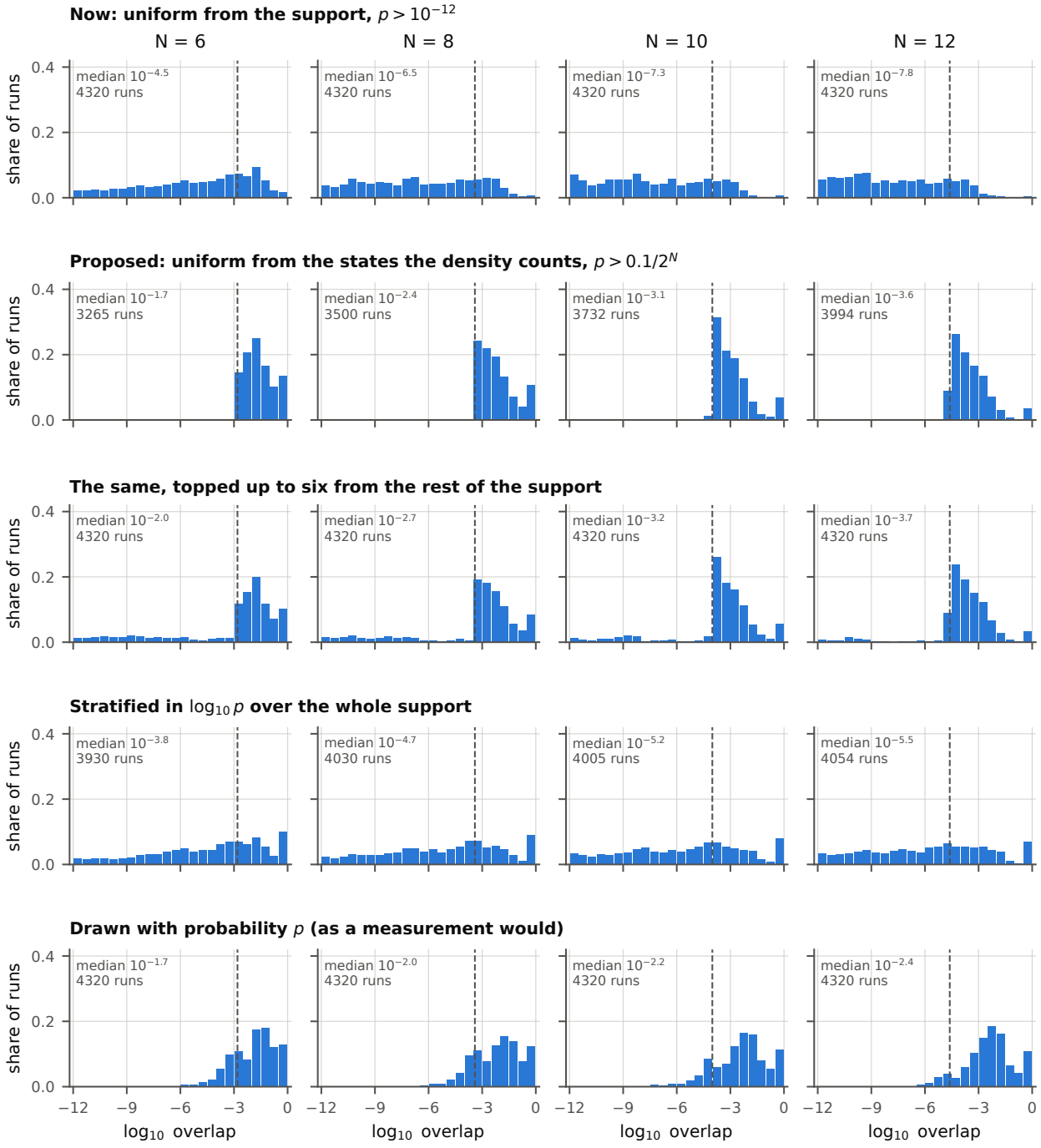


Figure 1: Distribution of the initial overlap per scheme (rows) and system size (columns), over all selected Hamiltonians. The dashed line is $0.1/2^N$, the threshold of the density.

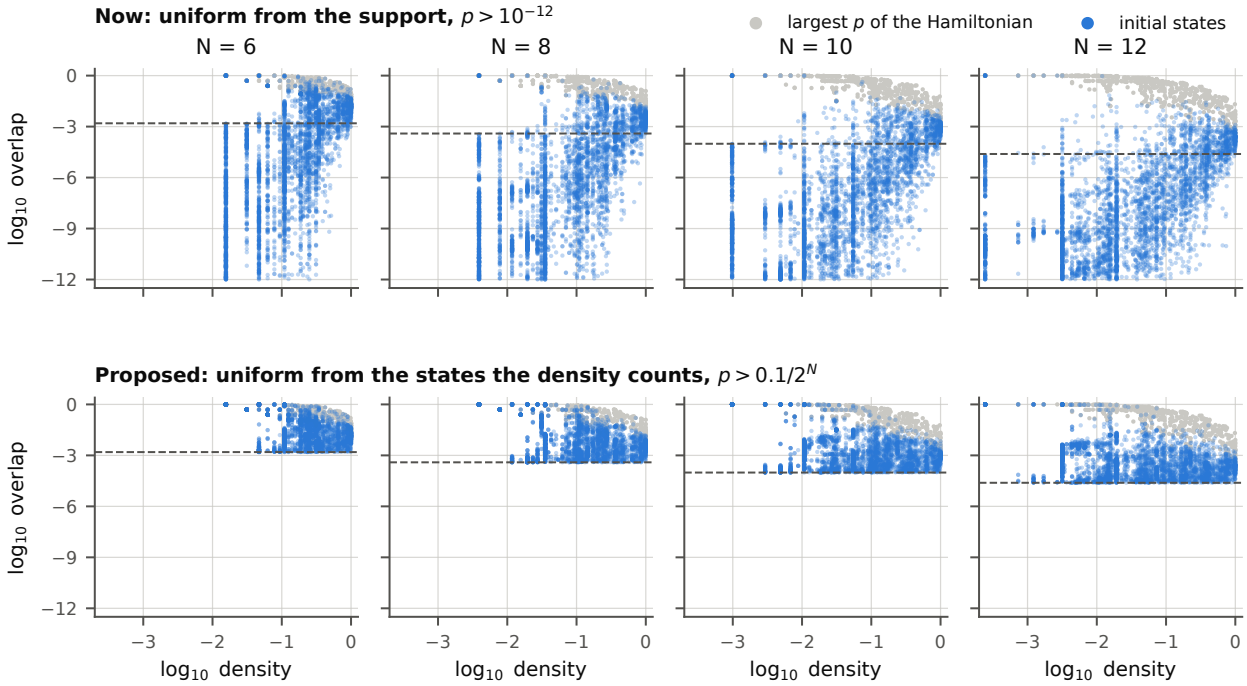


Figure 2: Initial overlap against ground-state density, now and with the recommended draw. Grey: the largest p of each Hamiltonian, i.e. the largest overlap it can offer at all. The dashed line is $0.1/2^N$.